

The Setup A Convention

What the “Setup A convention of Goudy and coworkers” means when it is cited for the protonic Fock operators of a multicomponent Hartree–Fock reference: the three setups defined in the source paper, what each includes, when each is to be used, why the choice changes perturbative corrections but not coupled cluster energies, and how a Hartree-product formulation realizes it.

Written for a graduate student who has read the NEO coefficients and NEO gradient notes

Definitions quoted from the supplementary material of Goudy, Pavošević and Hammes-Schiffer (2025)

NOTATION

i, j / I, J

electrons / quantum protons; α, β label distinguishable quantum nuclei where that distinction matters.

$\hat{h}^e, \hat{h}^p$

electronic and protonic core Hamiltonians, the latter containing the proton’s kinetic energy and its Coulomb interaction with the classical nuclei.

$v_{ee}^{\text{HF}}, v_{ep,e}^{\text{HF}}, v_{ep,p}^{\text{HF}}, v_{pp}^{\text{HF}}$

Fock potentials of the electrons on the electrons, the protons on the electrons, the electrons on the protons, and the protons on the protons.

$\hat{H}_0, \hat{V}$

zeroth-order Hamiltonian and perturbation in a Møller–Plesset-type partitioning, $\hat{H} = \hat{H}_0 + \hat{V}$.

$[A], [B], [C]$

superscripts marking the operators that differ between the three setups.

THE WHOLE ANSWER IN THREE LINES

- 1 Setup A keeps the proton–proton interaction in the Hamiltonian and its Hartree–Fock mean field in the protonic Fock operator, so the zeroth-order Hamiltonian is the sum of the full electronic and protonic Fock operators. It is the general form and the one required whenever more than one proton is quantum.
- 2 Setup B, for a single quantum proton, drops the proton–proton term from the Hamiltonian and from the Fock operator. Setup C keeps the Hamiltonian of A but moves the proton–proton mean field out of the zeroth-order operator, and is shown to be identical to B. NEO-HF, NEO-CCSD and NEO-CCSDT energies are the same in all three; virtual protonic orbitals and every perturbative correction are not.
- 3 For distinguishable protons the Setup A Fock operator of nucleus α is its core Hamiltonian, the electrons' mean field, and the Coulomb mean field of the other quantum nuclei, which is Eq. 3 of the multicomponent CCSD gradient paper.

1 Where the label appears

The analytic gradient paper for multicomponent CCSD by Tucker and Brorsen defines its reference in Section 2.1 with the sentence

“The reference is the multicomponent HF wave function with Fock operators defined using the Setup A convention of Goudy and coworkers. In this convention, the Fock operator for quantum nucleus α is”

KEY RESULT · THE SETUP A PROTONIC FOCK OPERATOR (EQ. 3 OF THE GRADIENT PAPER)

$$\hat{F}^{\text{p}\alpha} = \hat{h}^{\text{p}\alpha} + \hat{v}_{\text{HF}}^{\text{ep,p}\alpha} + \sum_{\beta \neq \alpha} \hat{v}_{\text{HF}}^{\text{p}\beta \text{p}\alpha}. \quad (1)$$

The citation is to Goudy, Pavošević and Hammes-Schiffer, *J. Chem. Phys.* **163**, 224119 (2025), the paper that introduced triple excitations into NEO coupled cluster theory for systems with several quantum protons. The gradient paper treats the quantum nuclei as distinguishable, with the product reference

$$|0\rangle = |0^e\rangle \otimes \prod_{\alpha} |0^{p\alpha}\rangle, \quad (2)$$

one occupied orbital and one orbital space per nucleus, and no exchange term in the proton–proton part of the two-particle density.

The Goudy paper defines the setups in Section S1 of its supplementary material, “Choice of the exact Hamiltonian and its partitioning for perturbation theory with a single quantum proton”. Its main text says what the distinction is about:

“The NEO-CCSD($T_{\text{eep},\text{epp}}$) energies were found to be sensitive to the inclusion or exclusion of the proton–proton interaction term in the Hamiltonian for systems containing only a single quantum proton. Although the contribution of this term to the NEO-HF energy is zero for systems with only a single quantum proton, its presence influences the virtual orbitals and the perturbative energy correction.”

2 The three setups, as defined

The supplementary material opens by noting that “multiple choices exist for how the exact Hamiltonian, the zeroth-order Hamiltonian, and the perturbation can be defined for perturbation theory involving a single quantum proton”, and considers three.

Setup A: the general Hamiltonian and its full Fock operators

Setup A “uses the most general form of the NEO Hamiltonian, which can be used for any number of quantum protons”,

$$\hat{H}_{\text{NEO}}^{[A]} = \sum_i \hat{h}^e(i) + \sum_I \hat{h}^p(I) + \sum_{i<j} r_{ij}^{-1} - \sum_{iI} r_{iI}^{-1} + \sum_{I<J} r_{IJ}^{-1}, \quad (3)$$

which leads to the Fock operators

$$\hat{F}^e = \sum_i [\hat{h}^e(i) + v_{\text{ee}}^{\text{HF}}(i) + v_{\text{ep},e}^{\text{HF}}(i)], \quad \hat{F}^{p[A]} = \sum_I [\hat{h}^p(I) + v_{\text{ep},p}^{\text{HF}}(I) + v_{\text{pp}}^{\text{HF}}(I)], \quad (4)$$

and to the partitioning

$$\begin{aligned}\hat{H}_0^{[A]} &= \hat{F}^e + \hat{F}^{p[A]}, & \hat{V}^{[A]} &= \hat{W}^{ee} + \hat{W}^{ep} + \hat{W}^{pp}, \\ \hat{W}^{pp} &= \sum_{I < J} r_{IJ}^{-1} - \sum_I v_{pp}^{\text{HF}}(I),\end{aligned}\tag{5}$$

with $\hat{W}^{ee}$ and $\hat{W}^{ep}$ the corresponding fluctuation potentials for the electron–electron and electron–proton interactions. The protons-on-protons Fock potential is in the zeroth-order operator, and the proton–proton fluctuation potential is in the perturbation.

Setup B: the proton–proton term removed from the Hamiltonian

“For a single quantum proton, the proton–proton interaction energy is zero, and therefore the proton–proton interaction term can be removed from the exact Hamiltonian”:

$$\hat{H}_{\text{NEO}}^{[B]} = \sum_i \hat{h}^e(i) + \sum_I \hat{h}^p(I) + \sum_{i < j} r_{ij}^{-1} - \sum_{iI} r_{iI}^{-1},\tag{6}$$

$$\hat{F}^{p[B]} = \sum_I [\hat{h}^p(I) + v_{ep,p}^{\text{HF}}(I)], \quad \hat{H}_0^{[B]} = \hat{F}^e + \hat{F}^{p[B]}, \quad \hat{V}^{[B]} = \hat{W}^{ee} + \hat{W}^{ep}.\tag{7}$$

The electronic Fock operator is unchanged. The protonic one loses its protons-on-protons potential, and the perturbation loses its proton–proton part.

Setup C: the Hamiltonian of A, repartitioned

Setup C keeps the Hamiltonian (3) and the Fock operator $\hat{F}^{p[C]} = \hat{F}^{p[A]}$ of Setup A, but moves the mean-field part of $\hat{W}^{pp}$ out of the zeroth-order operator:

$$\hat{H}_0^{[C]} = \hat{F}^e + \hat{F}^{p[C]} - \sum_I v_{pp}^{\text{HF}}(I), \quad \hat{V}^{[C]} = \hat{W}^{ee} + \hat{W}^{ep} + \sum_{I < J} r_{IJ}^{-1}.\tag{8}$$

Expanding $\hat{F}^{p[C]}$ shows that $\hat{H}_0^{[C]} = \hat{H}_0^{[B]}$, and because $\hat{V}$ is only ever evaluated between determinants that contain a single protonic orbital, the two-proton operator $\sum_{I < J} r_{IJ}^{-1}$ contributes nothing there and $\hat{V}^{[C]} = \hat{V}^{[B]}$. The supplementary material concludes that “the zeroth-order Hamiltonian and perturbation are exactly the same in setups [B] and [C]”, so the two give identical results “for any order of perturbation theory, and also for any perturbative coupled cluster methods that rely on this same partitioning”, and it discusses only Setup B from there on. Setup C exists to show that the earlier practice of removing $\sum_I v_{pp}^{\text{HF}}(I)$ from $\hat{H}_0$, which improved NEO-MP2, can be read either as a change of Hamiltonian or as a change of partitioning, with the same outcome.

SETUP	EXACT HAMILTONIAN	PROTONIC FOCK OPERATOR	ZEROth-ORDER HAMILTONIAN	PERTURBATION
A	general, with $\sum_{I<J} r_{IJ}^{-1}$	$\hat{h}^p + v_{\text{ep,p}}^{\text{HF}} + v_{\text{pp}}^{\text{HF}}$	$\hat{F}^e + \hat{F}^p[\text{A}]$	$\hat{W}^{\text{ee}} + \hat{W}^{\text{ep}} + \hat{W}^{\text{pp}}$
B	proton-proton term removed	$\hat{h}^p + v_{\text{ep,p}}^{\text{HF}}$	$\hat{F}^e + \hat{F}^p[\text{B}]$	$\hat{W}^{\text{ee}} + \hat{W}^{\text{ep}}$
C	same as A	same as A	same as B	same as B on the single-proton basis

3 When each setup is used

The supplementary material states the protocol directly. “When performing perturbative calculations involving multiple quantum protons, setup [A] is required.” With a single quantum proton the choice matters, and two considerations decide it.

- **Balance.** “When energies need to be compared for a calculation involving a single quantum proton and a calculation involving multiple quantum protons, the same setup should be used to maintain balance, indicating that setup [A] should be used.” This is why the proton affinities of methanol, methylamine, the water dimer and the protonated water dimer in the main text use Setup A even for their single-proton species. The supplementary material adds that mixing the setups loses size consistency: a two-proton system in Setup A does not match two one-proton systems in Setup B.
- **Accuracy.** “When energies do not need to be compared to energies of systems with multiple quantum protons, setup [B] can be used, providing an increase in accuracy.” Table S1 shows the effect for the single-proton protonated species: the mean unsigned error of the perturbative triples against the full treatment is 0.03 eV in Setup A and 0.01 eV in Setup B, while the NEO-CCSD and NEO-CCSDT_{eepp} values are identical in the two setups.

The deposited data on Zenodo labels the single-proton calculations accordingly: VPP1 for Setup A, “proton-proton interaction IS included”, and VPP0 for Setup B, with the Q-Chem keyword of the same name, and notes that for several quantum protons the interaction is always included.

4 Why the choice matters

For a single quantum proton the proton–proton term contributes nothing to the NEO-HF energy, and the occupied protonic orbital is unaffected. In a determinant-based formulation the protons-on-protons potential acting on the one proton is its own Coulomb minus exchange, which vanishes on the occupied orbital but not on the virtual space, so Setup A and Setup B have different virtual protonic orbitals and different orbital energies. The supplementary material also notes that B and C differ from A “not only [in] a different zeroth-order Hamiltonian”, which changes the basis in which corrections are evaluated, “but also a different definition of the perturbation”, and that “for NEO-MP2, it is strictly the change in the zeroth-order Hamiltonian that is impactful, but at higher orders of perturbation theory, the change in the perturbation can also be impactful”.

QUANTITY	DEPENDS ON THE SETUP	REASON, AND EVIDENCE IN THE SOURCE
NEO-HF energy	no	the proton–proton energy of a single proton is zero
Occupied protonic orbital	no	the self-terms cancel on it
Virtual protonic orbitals and energies	yes	the self-terms do not cancel on the virtual space
NEO-MP2	yes	through the zeroth-order Hamiltonian; the earlier NEO-MP2 work removed v_{pp}^{HF} for this reason
NEO-CCSD(T) and other perturbative triples	yes	through the zeroth-order Hamiltonian and the perturbation; Table S1 gives 0.03 eV versus 0.01 eV

QUANTITY	DEPENDS ON THE SETUP	REASON, AND EVIDENCE IN THE SOURCE
NEO-CCSD and NEO-CCSDT _{eep,epp} energies	no	“identical to those obtained when setup [B] is used”, Table S1; the iterated methods are invariant to the partitioning
Convergence of the amplitude iterations	yes	the denominators change with the orbital energies
Gradient Lagrangian pieces	yes	the Fock operators, their derivatives, the orbital Hessian blocks and the effective densities all carry the convention

The last row is why a gradient paper has to name the convention even though its correlated energy does not depend on it. The Lagrangian of the NEO gradient notes attaches the Brillouin conditions of each nucleus, differentiates each Fock matrix with respect to the perturbation, and builds the protonic orbital-Hessian blocks and the effective proton-proton density. Every one of these objects is different in a Fock operator with and without the protons-on-protons potential, and a gradient derived in one setup would be wrong in the other.

5 Setup A for distinguishable protons

Equation (1) is the Setup A Fock operator of (4) written for a Hartree product of distinguishable nuclei. The protons-on-protons potential acting on nucleus α is then the Coulomb mean field of the other nuclei, with no exchange, because distinguishable particles have none, and no self-term, because a Hartree product contains no exchange to cancel one. For several quantum protons, (1) and (4) agree up to the neglected inter-proton exchange and the self-terms, which cancel in the energy. The partitioning is Setup A’s: the proton-proton interaction stays in the Hamiltonian, its mean field sits in the zeroth-order operator, and its fluctuation potential sits in the perturbation and drives the proton-proton part of the correlation treatment.

For one quantum proton the sum in (1) is empty, so the Hartree-product Fock operator has no proton–proton term at all and coincides, as an operator, with $\hat{F}^{p[B]}$ of (7). The self-term $v_{pp}^{\text{HF}}(I)$ that distinguishes Setup A from Setup B in the determinant-based formulation of the Goudy paper never arises for distinguishable particles. So a Hartree-product code in the Setup A convention produces, for a single quantum proton, the protonic virtual orbitals of Setup B. Its NEO-CCSD energies match either setup, since those are invariant; its perturbative triples for a single-proton species match the Setup B column of Table S1, and would match a Setup A single-proton calculation only if the self-term were added back to the protonic Fock operator.

6 Practical consequences for codes

- **Obtaining Setup B from a Setup A implementation.** The supplementary material gives the recipe: exclude v_{pp}^{HF} from the nuclear Fock matrix during the NEO-HF procedure, then exclude all two-proton integrals in the CC calculation.
 - **Checking a code against the paper.** Compare NEO-CCSD energies first, since they are setup-independent. Then compare perturbative triples for a single-proton species against the Setup B results when the code is a Hartree product, or against Setup A results only after adding the self-term, as in the remark above. For species with several quantum protons, compare against Setup A, which is the only option there.
 - **Checking the NEO-HF layer.** Including or excluding the single-proton self-term must leave the NEO-HF energy unchanged to the convergence threshold and must shift the protonic virtual orbital energies; a code that fails either test has a bug in its proton–proton terms.
 - **Checking a gradient.** In Setup A for distinguishable protons the same-nucleus block of the protonic orbital Hessian reduces to orbital-energy differences and the coupling between nuclei is pure Coulomb; the effective proton–proton density has no exchange part. These are the structures the NEO gradient notes assume.
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7 Sources and what was verified

The quotation and Eq. (1) in Section 1 are taken from the open-access text of the gradient paper. Equations (3) to (8), the protocol of Section 3 and the quotations in Sections 2 to 4 are taken from Section S1 of the Goudy supplementary material, read in full, together with its Tables S1 and S2; the main text of the Goudy paper and its Zenodo deposit were read as well. The observation in Section 5 about single-proton systems is this document’s own, drawn by comparing Eq. (1) with Eqs. (4) and (7).

1. G. B. Tucker and K. R. Brorsen, “An Analytic Nuclear Energy Gradient for Multicomponent CCSD”, *J. Chem. Theory Comput.* **22**, 8813 (2026). Section 2.1, Eqs. (2) and (3), and reference 31.
2. R. J. Goudy, F. Pavošević and S. Hammes-Schiffer, “Triple excitations in nuclear–electronic orbital coupled cluster theory for multiple quantum protons”, *J. Chem. Phys.* **163**, 224119 (2025), and its supplementary material, Section S1 and Tables S1 and S2. The definitions of the setups and the protocol for their use.
3. R. J. Goudy, F. Pavošević and S. Hammes-Schiffer, data for the above, Zenodo, doi:10.5281/zenodo.17781752 (2025), CC BY 4.0. Inputs, outputs, the VPP0/VPP1 labels and the equation files.
4. C. Swalina, M. V. Pak and S. Hammes-Schiffer, “Alternative formulation of many-body perturbation theory for electron–proton correlation”, *Chem. Phys. Lett.* **404**, 394 (2005). The earlier removal of the protons-on-protons potential from the zeroth-order Hamiltonian at the NEO-MP2 level.
5. M. Chow, E. Paenurk and S. Hammes-Schiffer, “Efficient and robust optimization of nuclear and electronic orbitals within the nuclear Hartree product representation”, *J. Chem. Phys.* **164**, 074103 (2026). The Hartree-product and Slater-determinant protonic Fock operators and the cancellation of the self-terms.
6. F. Pavošević, T. Culpitt and S. Hammes-Schiffer, *Chem. Rev.* **120**, 4222 (2020). The NEO framework and NEO-HF.

Equation (1) is quoted from the gradient paper with its notation; Eqs. (3) to (8) follow the supplementary material of the Goudy paper, Eqs. (S1) to (S22), with its notation lightly condensed. Section 5 is this document’s own comparison of the two formulations.